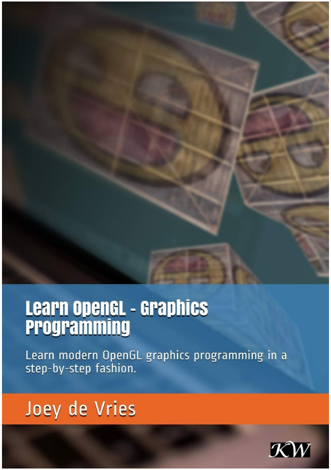
Hello Triangle

Introduction to Computer Graphics - Exercise 02

Pirmin Pfeifer

Resources

The OpenGL exercies are based on the LearnOpenGL course



Tools and Software

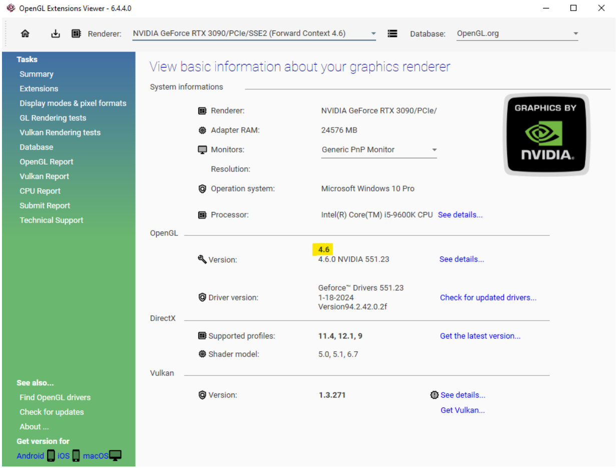
What you need:

- IDE for C++ (Visual Studio)
- CMake
- The provided base project from Moodle
- GPU with support for OpenGL Version 4.3.
  - dedicated GPU
  - modern integrated GPU (Intel HD Graphics or AMD Radeon)
- Git

Ensure the provided base project is compiling on your machine!

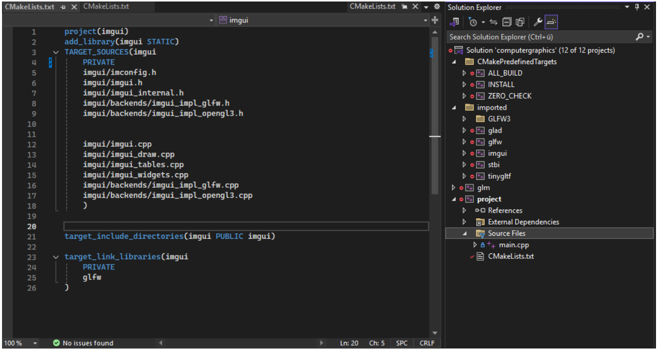
OpenGL Support

OpenGL Version 4.3. or above is recomended.
Check for your OpenGL support with **glxinfo** on Linux
or **OpenGL Extensions Viewer** on Windows.



The Project Structure

You can find a basic cmake project with the required libraries already setup on Moodle & Github.



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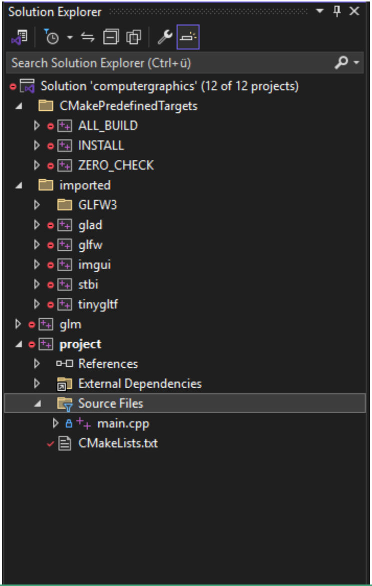

CmakePredefinedTargets:

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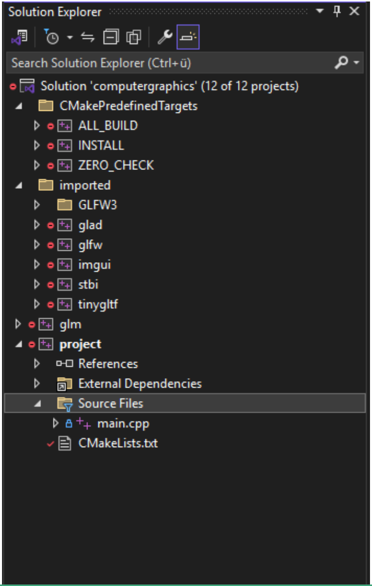

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- ALL_BUILD → start a build of the whole solution

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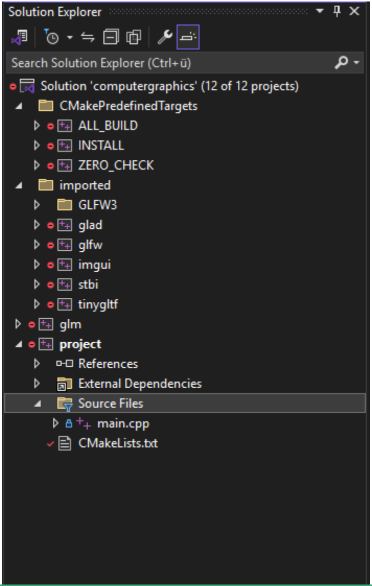

CmakePredefinedTargets:

- ALL_BUILD → start a build of the whole solution
- ZERO_CHECK → reruns cmake configure

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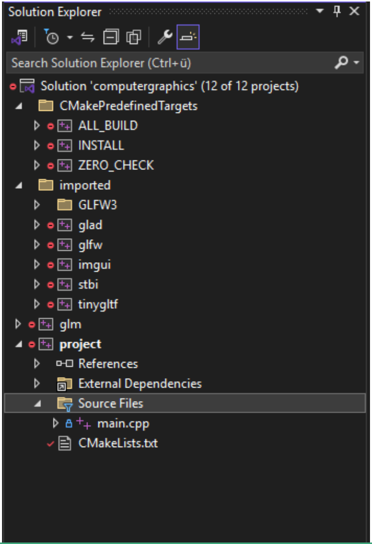
CMake is used to structure the Project



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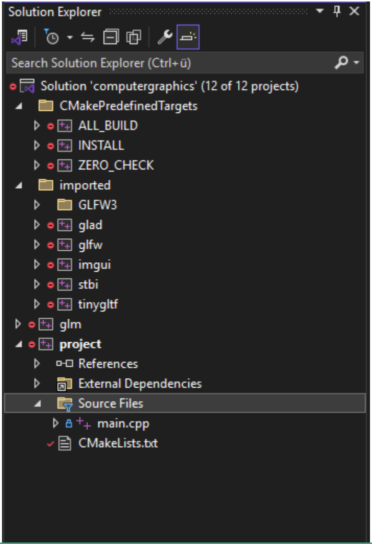

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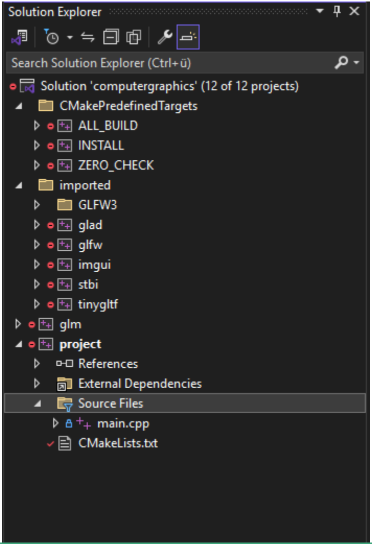

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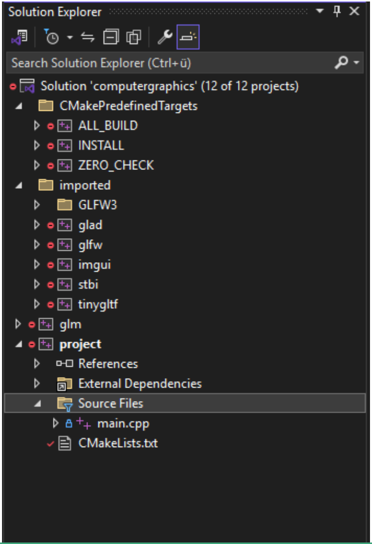

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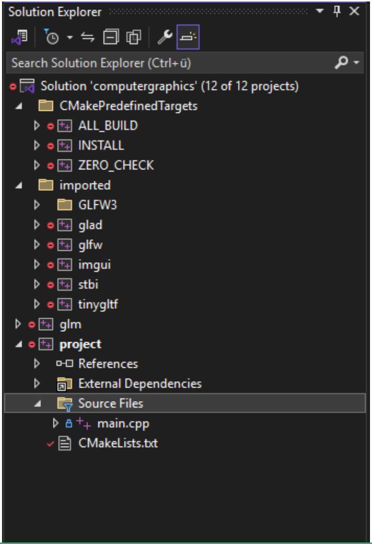

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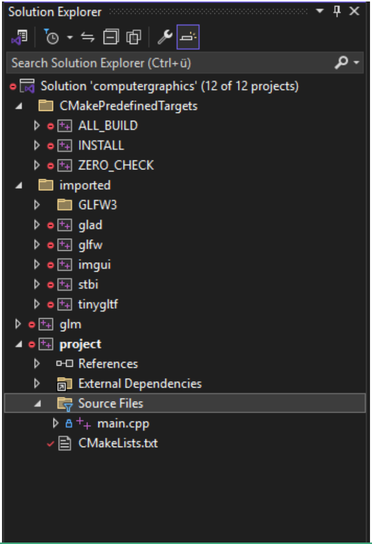

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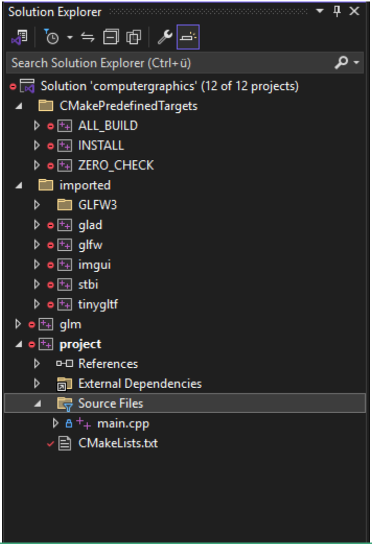

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What are the alternatives:

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Vulkan

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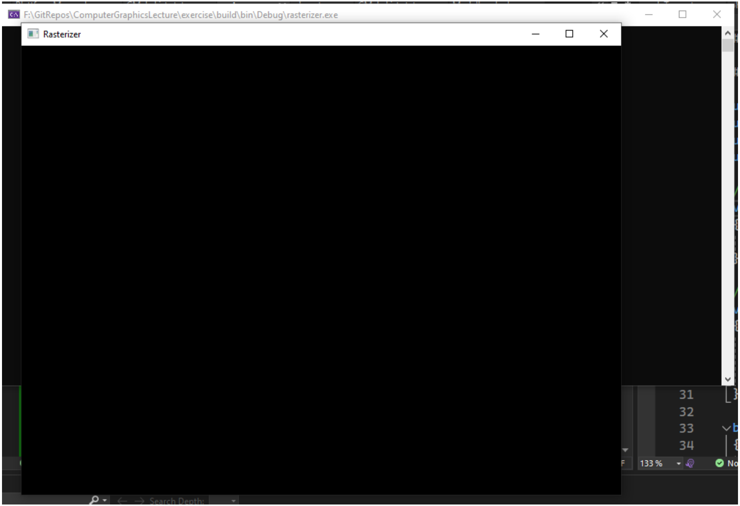
- even more LOC for a Triangle

Hello Window!

Lets start with creating a Window:



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Hello Window!

What do we need[1]:

- glad - OpenGL Context & Function bindings
- GLFW - Platform independent Window abstractions

```cpp
// glad for OPGL context
#include <glad/glad.h>
// GLFW for window management
#include <GLFW/glfw3.h>
```

File: main.cpp

[1] The base project should compile and the compatibilityCheck should run with out error

Hello Window! - Steps

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7. Add Render & Update Loop
8. Clean up after loop

Hello Window! - Render Loop

The render loop is used to update the image displayed in the window:

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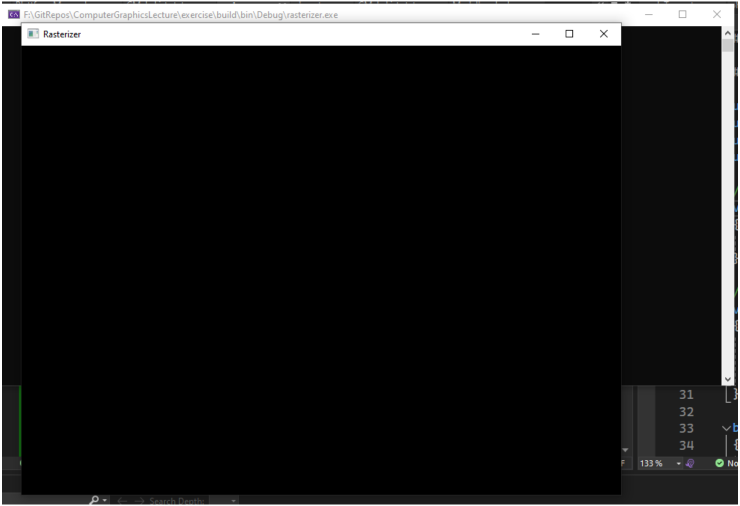
The render loop is used to update the image displayed in the window:

1. Handle Input
2. Clear the canvas/framebuffer
3. Render new image (at first this will just be setting a flat color)
4. Swap buffers to present the rendered image
5. pool events to recive updates such as key down event or resizes

Hello Window!

Now we have a nice black window:



Hello Input! Bye Window!

Close on escape key pressed:

```cpp
while (!glfwWindowShouldClose(window))
{
    if(glfwGetKey(window, GLFW_KEY_ESCAPE) == GLFW_PRESS) {
        glfwSetWindowShouldClose(window, true);
    }

    glfwSwapBuffers(window);
    glfwPollEvents();
}
```

File: main.cpp (render loop)

Hello Color!

In the render loop we can clear the buffer using a ClearColor:

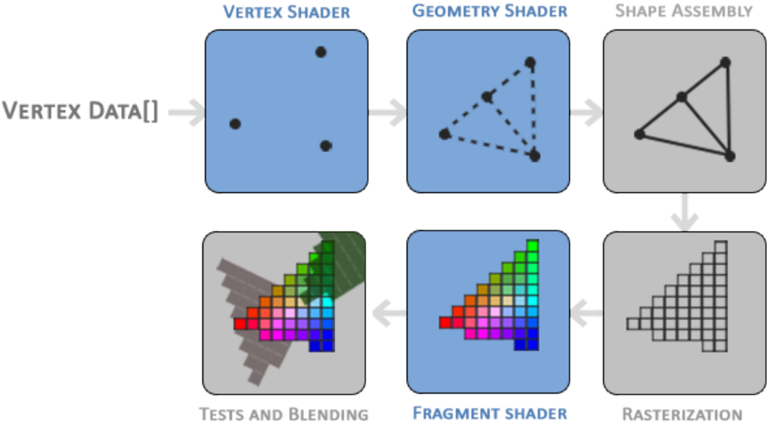
```
glClearColor(0.2f, 0.3f, 0.3f, 1.0f);
glClear(GL_COLOR_BUFFER_BIT);
```

File: main.cpp (updated render loop)

Now it is your turn!

Use LearnOpenGL to create a window!

Hello Triangle!



Hello Triangle! - Vertex Input

Vertex Coordinates in a array:

```cpp
float vertices[] = {
    -0.5f, -0.5f, 0.0f,
     0.5f, -0.5f, 0.0f,
     0.0f,  0.5f, 0.0f
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File: main.cpp

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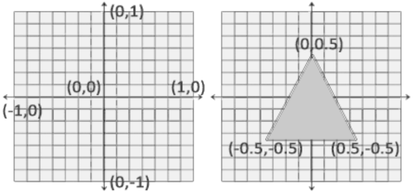
File: main.cpp Normalized Device Coordinates (NDC):



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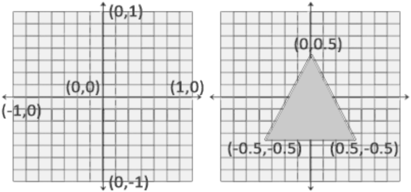

NCD will be transformed into view space coordinates using the specified viewport

Hello Triangle! - Vertex Input

We need to upload our vertices to the GPU before we can render them:

File: main.cpp

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```cpp
// Generate a Vertex Buffer Object
unsigned int VBO;
glGenBuffers(1, &VBO);
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glBindBuffer(GL_ARRAY_BUFFER, VBO);
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glBindBuffer(GL_ARRAY_BUFFER, VBO);
```

3. Upload data to buffer

```
glBufferData(
    GL_ARRAY_BUFFER, sizeof(vertices), vertices, GL_STATIC_DRAW);
```

File: main.cpp

Hello Triangle! - Vertex Shader

```glsl
#version 330 core
layout (location = 0) in vec3 aPos;

void main()
{
    gl_Position = vec4(aPos.x, aPos.y, aPos.z, 1.0);
}
```

Vertex Shader - Load and Compile

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Compile the shader source:

```
const char* vertexShaderSourcePtr = vertexShaderSource.c_str();
glShaderSource(vertexShader, 1, &vertexShaderSourcePtr, NULL);
glCompileShader(vertexShader);
```

File: main.cpp

Vertex Shader - Error Checking

We probably should check if the compilation was successful:

```cpp
int  success;
char infoLog[512];
glGetShaderiv(vertexShader, GL_COMPILE_STATUS, &success);
if(!success)
{
    glGetShaderInfoLog(vertexShader, 512, NULL, infoLog);
    std::cout << "ERROR::SHADER::VERTEX::COMPILATION_FAILED\n"
              << infoLog
              << std::endl;
    return -1;
}
```

File: main.cpp

Fragment Shader

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```
#version 330 core
out vec4 FragColor;

void main()
{
    FragColor = vec4(1.0f, 0.5f, 0.2f, 1.0f);
}
```

Fragment Shader - Load and Compile

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Error checking is the same as with the vertex shader.
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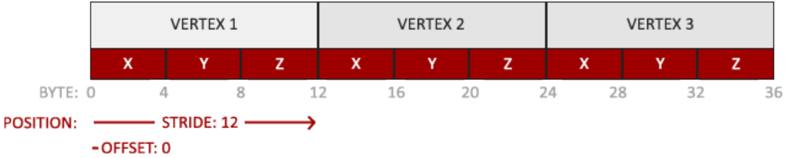
At the end of our program we should release the shaders with `glDeleteShader( shaderId )`.

File: main.cpp

Linking Vertex Attributes

Lets have another look at our vertex buffer:



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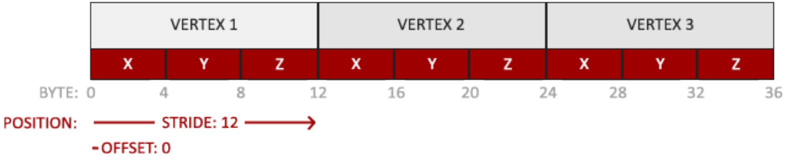

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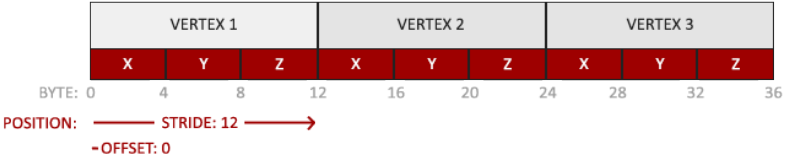

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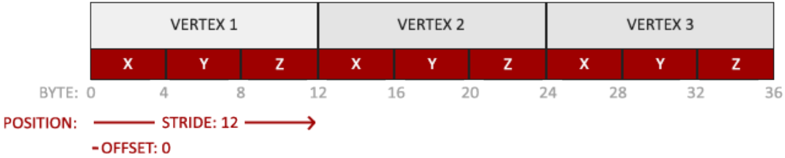

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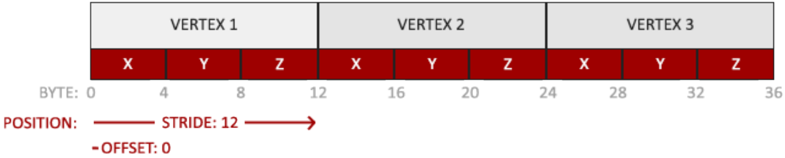

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- The first value in the data is at the beginning of the buffer.

Vertex Array Object

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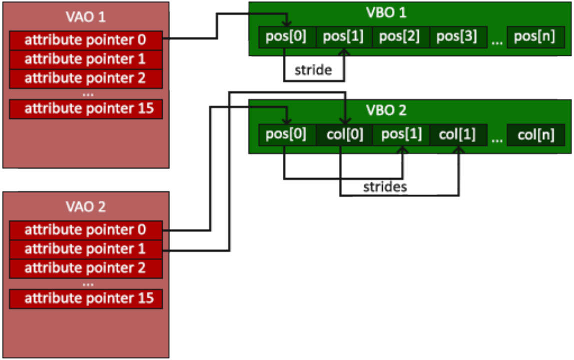
Vertex Array Object

We need to tell the GPU where in our buffer the vertices are and how they are aligned.



We use a Vertex Array Object to provide the Vertex Attributes Information.

Vertex Array Object

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unsigned int VAO;
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glBindVertexArray(VAO);
// 2. copy our vertices array in a buffer for OpenGL to use
glBindBuffer(GL_ARRAY_BUFFER, VBO);
glBufferData(
    GL_ARRAY_BUFFER, sizeof(vertices), vertices, GL_STATIC_DRAW);
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[2] we only need to do this once, but we need to bind the buffer here so we can combine both

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Set up the Vertex Attribute Pointer:

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```

Set up the Vertex Attribute Pointer:

```
// 3. then set our vertex attributes pointers
glVertexAttribPointer(
    0, 3, GL_FLOAT, GL_FALSE, 3 * sizeof(float), (void*)0);
glEnableVertexAttribArray(0);
```

File: main.cpp

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Vertex Array Object

Lets have a closer look at: `glVertexAttribPointer (`

```
)
```

```
glVertexAttribPointer (  ,  ,           ,           ,                ,        );
```

Vertex Array Object

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Lets have a closer look at: `glVertexAttribPointer (`

- `GLuint index` index of the attribute in the shader

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Vertex Array Object

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Vertex Array Object

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- `GLsizei stride` byte offset between consecutive vertex attributes
- `const GLvoid * pointer` offset of the first component in the array

`)`

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glUseProgram(shaderProgram);
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1. bind the shader program before the render loop

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glUseProgram(shaderProgram);
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glBindVertexArray(VAO);
```
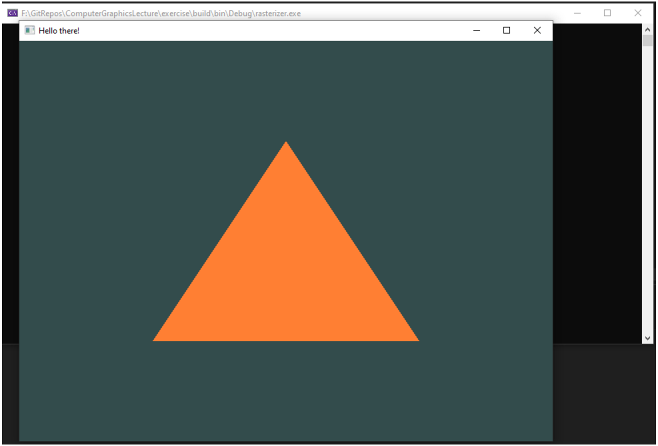

3. make our first draw call

```
glDrawArrays(GL_TRIANGLES, 0, 3);
```

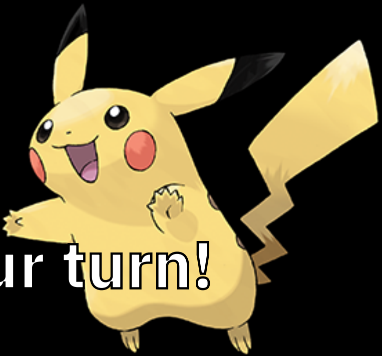
File: main.cpp

Hello There!

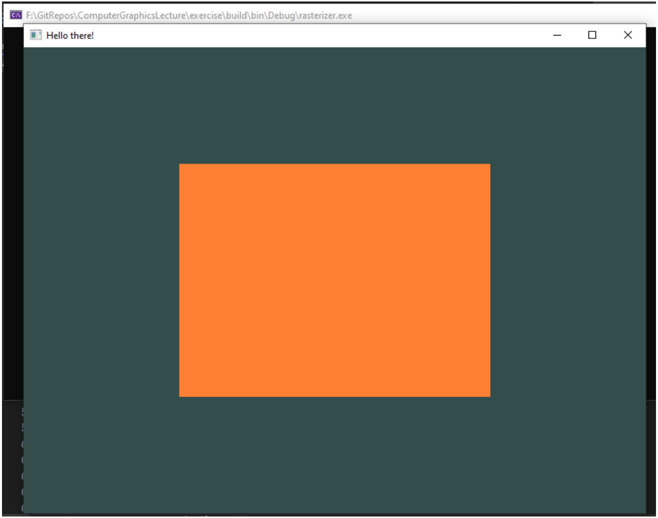


F:\GitRepos\ComputerGraphicsLecture\exercise\build\bin\Debug\rasterizer.exe
Hello there!

Now it is your turn!



Let’s think bigger!



Let’s think bigger!

Lets look at our vertex buffer for the quad:

```cpp
float vertices[] = {
    // first triangle
    0.5f,  0.5f, 0.0f,  // top right
    0.5f, -0.5f, 0.0f,  // bottom right
    -0.5f,  0.5f, 0.0f,  // top left
    // second triangle
    0.5f, -0.5f, 0.0f,  // bottom right
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File: main.cpp

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File: main.cpp

Let's think smaller!

We can split our geometry info into vertex and index info:

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};
unsigned int indices[] = {  // note that we start from 0!
    0, 1, 3,   // first triangle
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File: main.cpp

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We reduced our memory cost:

$$4b * 3c * 6v = 72 bytes \Rightarrow ?b * ?c * ?v + ?b * ?c * ?t = ??bytes$$

b :=bytes, c :=components , v :=vertices, t :=triangles

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Which is nothing at the moment, but we generally add more vertex attributes.

Index Buffer

Allocate Element Buffer Object (index buffer):

```
unsigned int EBO;
glGenBuffers(1, &EBO);
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Bind and upload buffer:

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glBindBuffer(GL_ELEMENT_ARRAY_BUFFER, EBO);
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             sizeof(indices), indices,
             GL_STATIC_DRAW);
```

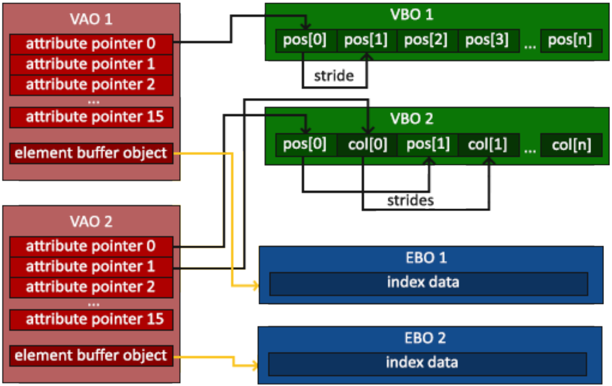
In render loop we now also need to bind the index buffer and use a different draw call:

```
glBindBuffer(GL_ELEMENT_ARRAY_BUFFER, EBO);
glDrawElements(GL_TRIANGLES, 6, GL_UNSIGNED_INT, 0);
```

File: main.cpp

Index Buffer

The Element Buffer Object is also bound into the Vertex Array Object:



Index Buffer

Our Vertex Array Object inti code look now like this:

```
// 1. bind Vertex Array Object
glBindVertexArray(VAO);
// 2. copy our vertices array in a vertex buffer for OpenGL to use
glBindBuffer(GL_ARRAY_BUFFER, VBO);
glBufferData(GL_ARRAY_BUFFER,
             sizeof(vertices), vertices,
             GL_STATIC_DRAW);
// 3. copy our index array in a element buffer for OpenGL to use
glBindBuffer(GL_ELEMENT_ARRAY_BUFFER, EBO);
glBufferData(GL_ELEMENT_ARRAY_BUFFER, sizeof(indices),
             indices,
             GL_STATIC_DRAW);
// 4. then set the vertex attributes pointers
glVertexAttribPointer(
    0, 3, GL_FLOAT, GL_FALSE, 3 * sizeof(float), (void*)0);
glEnableVertexAttribArray(0);
```

File: main.cpp

Index Buffer

Finally we update our render loop to something like this:

```
glUseProgram(shaderProgram);
while (!glfwWindowShouldClose(window)) {
    // input
    processInput(window);

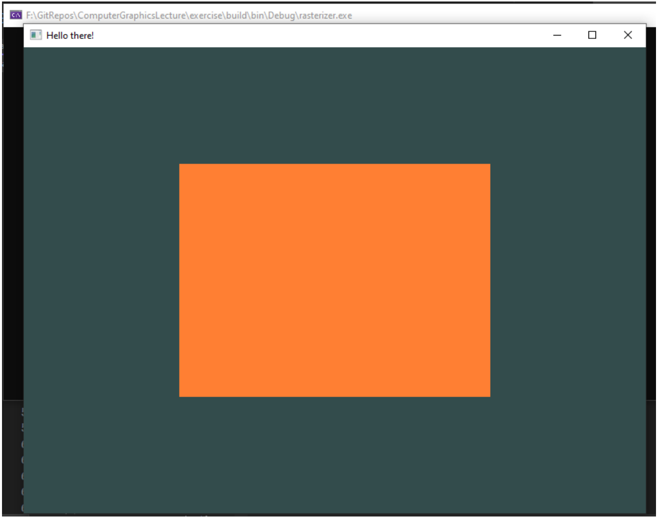
    // render
    glClearColor(0.2f, 0.3f, 0.3f, 1.0f);
    glClear(GL_COLOR_BUFFER_BIT);

    glBindVertexArray(VAO);
    glDrawElements(GL_TRIANGLES, 6, GL_UNSIGNED_INT, 0);

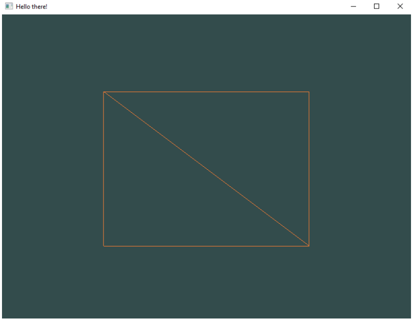
    // present frame and update events
    glfwSwapBuffers(window);
    glfwPollEvents();
}
```

Index Buffer

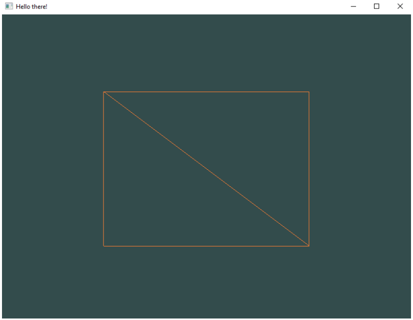
Now we have the same quad, but rendered with an index buffer:



Bonus: Wireframe



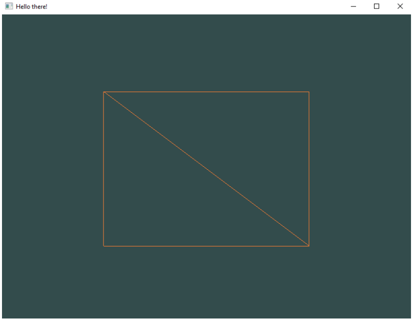
Bonus: Wireframe



We can enable Wireframe mode with:

```
glPolygonMode(GL_FRONT_AND_BACK, GL_LINE)
```

Bonus: Wireframe



File: main.cpp (render loop)

We can enable Wireframe mode with:

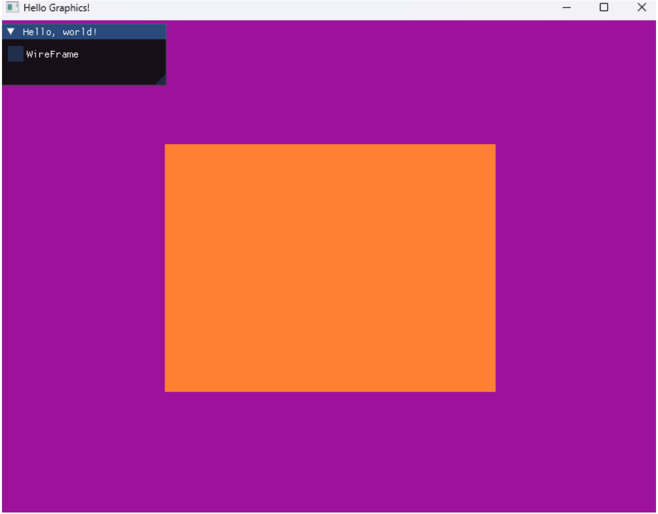
```
glPolygonMode ( GL_FRONT_AND_BACK , GL_LINE )
```

We can disable it again with:

```
glPolygonMode ( GL_FRONT_AND_BACK , GL_FILL )
```

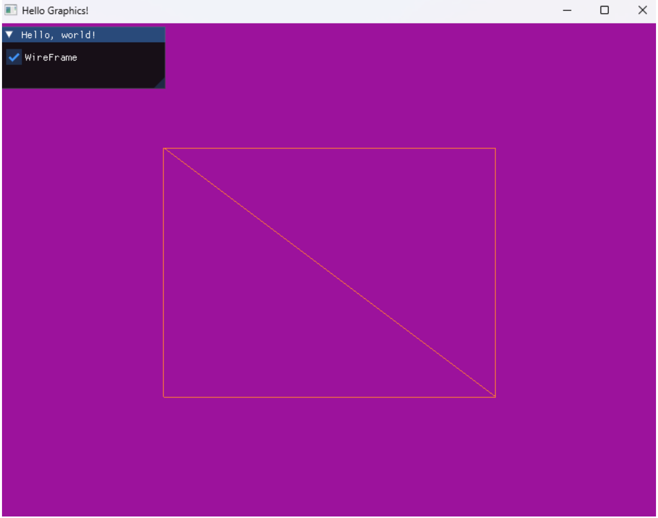
Bonus: Wireframe

We can create a toggle for the wireframe with ImGui:



Bonus: Wireframe

We can create a toggle for the wireframe with ImGui:



Bonus: ImGui

Requiered Includes:

```
#include "imgui.h"
#include "backends/imgui_impl_glfw.h"
#include "backends/imgui_impl_opengl3.h"
```

File: main.cpp

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```

To use ImGui we need to setup imgui with opengl:

```
// Setup Dear ImGui context
IMGUI_CHECKVERSION();
ImGui::CreateContext();
ImGuiIO& io = ImGui::GetIO();
io.ConfigFlags |= ImGuiConfigFlags_NavEnableKeyboard;
io.ConfigFlags |= ImGuiConfigFlags_NavEnableGamepad;

// Setup Platform/Renderer backends
ImGui_ImplGlfw_InitForOpenGL(window, true);
ImGui_ImplOpenGL3_Init();
```

File: main.cpp

Bonus: ImGui

Inside the render loop we need to prepere the next frame for ImGui:

```
ImGui_ImplOpenGL3_NewFrame();
ImGui_ImplGlfw_NewFrame();
ImGui::NewFrame();
```

File: main.cpp

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```
ImGui_ImplOpenGL3_NewFrame();
ImGui_ImplGlfw_NewFrame();
ImGui::NewFrame();
```

Afterwards we can create our UI Logic:

```
ImGui::Begin("Hello, world!");
ImGui::SetWindowSize(ImVec2(200, 75), ImGuiCond_Once);
ImGui::Checkbox("WireFrame", &wireframe);
ImGui::End();
```

File: main.cpp

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```

At the end, but before swaping the buffers, issue the ImGui Rendering:

```
ImGui::Render();
ImGui_ImplOpenGL3_RenderDrawData(ImGui::GetDrawData());
```

File: main.cpp

Now it is your turn again!